[INCH-POUND]
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

STUD WELDING SYSTEMS, INTEGRAL POWER SOURCE AND CONTROL UNIT, MANUAL

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers electric arc and capacitor discharge stud welding systems with integral direct current power source, controller, and stud gun. The stud welding systems are intended for manual welding use in operations that require the rapid, reliable, and economic attachment of stud fasteners or bolts to a plate or panel.

2. CLASSIFICATION. The stud welding systems shall be of the following types and sizes. The type and size to be furnished shall be as specified (see 7.2(b)).

Type I - Capacitor discharge (CD)

Size 1 - 12 gauge through 1/4-inch
Size 2 - 12 gauge through 5/16-inch
Size 3 - 1/8-inch through 3/8-inch

Type II - Electric arc

Size 4 - 10 gauge through 1/2-inch
Size 5 - 10 gauge through 5/8-inch
Size 6 - 10 gauge through 3/4-inch
Size 7 - 10 gauge through 7/8-inch
Size 8 - 5/16-inch through 1-inch
Size 9 - 10 gauge through 1-1/4-inch

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3431

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3. SALIENT CHARACTERISTICS

3.1 General requirements. The system shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustments, replacement, and repair.

3.2 Components. The stud welding shall include, as a minimum, the following components:

3.2.1 Power source and controller (PSC).

3.2.1.1 Type I, stud welder. The PSC for type I stud welding systems shall contain the capacitor bank and controls for power on/off, capacitance range, and charging voltage. The PSC shall possess jacks for the attachment of welding cables. The systems shall be initial contact, drawn arc, or initial gap systems as specified (see 7.2(c)). The PSC shall automatically discharge after the power switch is placed in the "off" position and shall have a protective device to prevent the capacitor voltage from being applied to the stud gun until the gun is properly seated on the workpiece.

3.2.1.2 Type II, stud welder. The II stud welding systems shall operate on the standard electric drawn arc principle. The PSC shall contain the transformer-rectifier and controls for power on/off, welding current, and timing. The PSC shall possess connectors for the attachment of welding cables. When specified (see 7.2(d)), the systems shall be capable of reverse polarity welding.

3.2.2 Stud welding gun. The stud welding gun shall be constructed of nonconductive, high-impact material and shall be capable of withstanding the rigorous handling normally encountered in construction environments. The gun shall be of the pistol type with a finger-actuated switch that initiates the welding process.

3.2.2.1 Stud welding gun adjustements. Type I and type II stud welding guns shall be equipped with adjustable support legs and stud chucks as specified (see 7.2(e)) to accommodate desired stud lengths and diameters. Type II systems shall be equipped with stud gun feet and arc shield ferrule grips as specified (see 7.2(f)).

3.2.2.2 Dual-gun system. When specified (see 7.2(g)), type II stud welders shall have a dual-gun system.

3.2.3 Welding cables. The stud welder shall possess a welding cable assembly to provide a means for conducting welding current and transmitting control signals from the welding gun to the controller. A means shall be provided to securely clamp the ground lead(s) to the workpiece. Cable lengths shall be provided as specified (see 7.2(h)).

3.2.4 Inert gas shielding equipment. If required, the type II stud welding system shall be equipped with gas shielding accessories as specified (see 7.2(i)).

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3.2.5 Portability. Type I stud welding systems shall be manually portable. Type II systems shall be manually portable, or the base shall be equipped with wheels or with lifting eye(s) as specified (see 7.2(j)).

3.2.6 Electrical system. The electrical system shall conform to the American National Standards Institute/National Fire Protection Association (ANSI/NFPA) 79, "Machinery, Industrial, Electrical Standard For". Unless otherwise specified (see 7.2(k)), type II stud welding systems shall operate from a dual 230/460-volt, 3-phase, 60-hertz circuit and shall be wired for the voltage. Type I systems shall operate from a 120-volt, single-phase, 60-hertz circuit. Type II electrical systems shall provide current output regulation and thermal overload protection as specified (see 7.2(l)).

3.2.7 Optional accessories. Optional accessories shall be furnished as specified (see 7.2(m)).

3.3 Performance. Unless otherwise specified (see 7.2(n)), the stud welding system shall meet the performance requirements in sections 3.2.1 and 3.2.2 when tested in accordance with sections 5.3 through 5.3.3 herein.

3.3.1 Visual weld inspection. For the type I process, the fillet formed at the base of the stud shall appear full and even. The weld shall not form a large crater with excessive metal expulsion or possess a very shiny appearance. For the type II process, the fillet shall appear full, even, and shiny all around the stud. The fillet shall not possess a very shiny, low profile extending beyond the outside of the ferrule, nor shall it possess a small, uneven, or dull appearance with fingers of metal extending through the vents of the ferrule.

3.3.2 Weld strength. A perpendicular force with sufficient magnitude to bend the stud shall not cause failure or cracking in the weld. Failure can occur in the stud or the workpiece.

3.4 Machine dimensions and weight. If required, the maximum dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (7.2(o)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI/National Electrical Manufacturers Association (ANSI/NEMA) EW1, "Welding, Electric Arc, Power Sources", and ANSI/American Welding Society (ANSI/AWS) C5.4, "Stud Welding, Recommended Practices For". The manufacturer shall ensure that the stud welding system and all equipment and accessories supplied with the system shall be in compliance with ANSI/AWS Z49.1, "Safety In Welding And Cutting", and Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910, "Occupational Safety and Health Standards for General Industry". If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply. Covers, guards, or other safety devices shall be provided for all parts of the system that present safety hazards. The manufacturer shall ensure that the system is intended for manual use. A type II system having a power source with an open circuit voltage that is not in accordance with NEMA EW1 section 5.3.1 shall not be provided as a manual system. Should this situation occur, it shall be cause for rejection of the system.

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3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(p)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(q)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(r)), the nameplate shall contain the following information:

a. Manufacturer's name
b. System model designation
c. Power input (volts, total amperes, phase, frequency)
d. Rated load volts and amperes
e. Maximum open circuit voltage
f. Duty cycle at rated load
g. Contract number or order number
h. National stock number
i. Date of manufacture

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same system offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Inspection. The stud welding system shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(s)), the stud welding system shall be tested in accordance with sections 5.3.1 through 5.3.3.

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5.3.1 Test conditions. The stud welding system shall be tested in an indoor facility under the ambient temperature and humidity conditions that normally occur within the facility.

5.3.2 Operational test. The stud welding system shall be operated in accordance with the manufacturer's standard operating procedures. The system shall be operated at no load for not less than 30 minutes. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. The performance tests shall consist of welding 10 round studs of the smallest diameter for the size machine under test. The procedure shall be repeated for 10 studs of the maximum diameter for the size machine under test. The workpieces for the tests shall be flat plates with a minimum thickness 1/3 the diameter of the test studs. The workpieces and studs shall consist of C-1008 to C-1022 steel. The welded studs shall be inspected visually to determine compliance with the requirements of section 3.3.1. Each welded stud shall undergo a bend test whereby a hammer or pipe is used to bend the stud to a minimum of 45 degrees from vertical. The welded studs shall comply with the requirements of section 3.3.2.

5.4 Acceptance. Unless otherwise specified (see 7.2(t)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the performance tests shall be compared to the performance requirements of sections 3.3.1 and 3.3.2. Failure of the system to meet the performance requirements shall be cause for rejection.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the products shall be preserved, packed, and marked as specified in the acquisition order (see 7.2(u)).

7. NOTES

7.1 Sources of documents.

7.1.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

7.1.2 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

7.1.3 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

7.1.4 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

7.1.5 OSHA standards. Copies of OSHA standards may be obtained from the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

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7.2 Ordering data. Acquisition documents must specify the following information:

a. Title, number, and date of this document
b. Type and size of stud welding system required (see 2)
c. Specify initial contact, drawn arc, or initial gap, as required (see 3.2.1.1)
d. Reverse polarity capability, if required (see 3.2.1.2)
e. Stud gun legs and chucks, as specified (see 3.2.2.1)
f. Stud gun foot and ferrule grip, as specified (see 3.2.2.1)
g. Dual-gun system, if required (see 3.2.2.2)
h. Specify cable length (see 3.2.3)
i. Gas shielding accessories, if required (see 3.2.4)
j. Wheels or lifting eye(s), as specified (see 3.2.5)
k. Electrical requirements, if different (see 3.2.6)
l. Type II line regulation, current output regulation, thermal overload protection, as specified (see 3.2.6)
m. Optional equipment, as specified (see 3.2.7)
n. Performance, if different (see 3.3)
o. Machine dimensions and weight, if required (see 3.4)
p. System of units required (see 3.6)
q. Dual system of units, if required (see 3.6)
r. Nameplate, if different (see 3.7)
s. Testing, if different (see 5.3)
t. Acceptance, if different (see 5.4)
u. Packaging requirements (see 6)

7.2 Part identification number (PIN). The following part identification numbering system is for government purposes and does not constitute a requirement for the contractor.

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- Size: 1 - 12 gauge through 1/4-inch; 2 - 12 gauge through 5/16-inch; 3 - 1/8-inch through 3/8-inch; 4 - 10 gauge through 1/2-inch; 5 - 10 gauge through 5/8-inch; 6 - 10 gauge through 3/4-inch; 7 - 10 gauge through 7/8-inch; 8 - 5/16-inch through 1-inch; 9 - 10 gauge through 1-1/4-inch
- Type: A - Type I, Capacitor discharge (CD); B - Type II, Electric arc
- CID number

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MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - AL
Navy - SH

Reviewers:
Air Force - 84
Army - AV, CE
Navy - AS, CG

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

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